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A perspective on friction interventions to curb the spread of misinformation



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Social media has enabled the spread of information at unprecedented speeds and scales, and with it the proliferation of high-engagement, low-quality content. *Friction*—behavioral design measures that make the sharing of content more cumbersome—might be a way to raise the quality of what is spread online. In this perspective, we propose a scalable field experiment to study the effects of friction with a learning component to educate users on the platform’s community standards. Preliminary simulations from an agent-based model suggest that while friction alone may decrease the number of posts without improving their quality, it could significantly increase the average quality of posts when combined with learning. The model also suggests that too much friction could be counterproductive. Experimental interventions inspired by these findings would be minimally invasive.

The spread of misinformation online has been recognized as a global societal threat to democracy, eroding trust in mainstream news sources, authorities, experts, and other socio-political institutions^{1–4}. Social media platforms have enabled the sharing of information at unprecedented speeds and scales, and with it came the proliferation of not only misinformation, but also other low-quality and harmful content such as hate speech, cyberbullying, and malware^{2,5}. Large social media platforms amplify the so-called attention economy, where information in abundance competes for scarce attention^{1,6,7}. Through this competition, one would hope for accurate information to surface from the interactions among many users by combining *independent* opinions in accordance with *wisdom of crowds* effects⁸. Alas, scholars have demonstrated that engaging yet false content gets shared more and travels faster than correct information⁹.

Socio-cognitive biases and algorithmic sorting both contribute to the spreading of high-engagement over high-quality content. Classic strategies such as accepting new information if it comes from multiple sources¹⁰ or through posts that have been shared many times fail because the aggregate opinions to which we are exposed online are not necessarily independent^{11,12}. This trait is boosted by confirmation bias¹³, a disconnect between what users deem accurate and what they deem shareable^{14,15}, and habits to share the most engaging content¹⁶. The illusory truth effect¹⁷ exacerbates vulnerability to misinformation in social media by increasing the perceived truth value of low-quality content through repetition^{18,19}.

Biases in content sorting algorithms of social media platforms also prioritize high engagement²⁰, increasing the exposure of low-quality content in user news feeds². One-click reactions, such as “Like” or “Share,” are driving mechanisms behind algorithmic sorting as they are easy to use and quickly influence the popularity of posts. Such user reactions are typically

presented in aggregate form as a popularity metric beneath the post, steering user attention. For example, a high retweet count is likely to be perceived as a crowd-sourced trust signal^{8,21}, possibly contributing to the content’s virality²² irrespective of quality²³. One corollary of these dynamics is an incentive for influence operations based on inauthentic behaviors, such as coordinated liking^{24–32}.

Scholars have called for ways to promote the Internet’s potential to strengthen rather than diminish democratic virtues and public debate³³ and to leverage the economics of information for protection rather than for misguidance³⁴. A relatively recent idea is to improve the quality of what is shared online by introducing *friction* on social media. The hope is to curb the spread of harmful content and misinformation by making it more difficult to share or like content online^{1,34,35}.

The focal question of this perspective is whether adding a bit of *friction* to the sharing process might mitigate the spread of low-quality and malicious content. We propose a field experiment to explore how friction may positively affect information quality in a social media environment. We are particularly interested in friction intervention with a quality-recognition learning component. Learning would take place by friction prompts that quiz users about a platform’s community standards.

To study the potential impact of the proposed friction intervention, we carry out a proof-of-concept study using a model of information sharing on social media. Such models may indeed aid in intervention design. For example, one mathematical model suggests that friction may be beneficial when applied to mass-sharing³⁶. The model assumes that messages mutate at every instance of re-sharing (deliberately or inadvertently). It shows that caps on depth (how many times messages may be forwarded) or breadth (the number of others to whom messages can be forwarded) improve the ratio of true to false messages.

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We present preliminary results from an agent-based model (ABM) to evaluate a simplified version of the proposed field experiment. Our simulations suggest that friction alone decreases the number of posts without improving their average quality. On the other hand, a small amount of friction combined with quality-recognition learning increases the average quality of posts significantly.

Friction and learning interventions

In the context of online interactions, *behavioral friction*, in general, denotes “any unnecessary retardation of a process that delays the user accomplishing a desired action”³⁷. The more friction, the lower the chances that the user will complete the action. While reducing friction is generally deemed desirable in user interface design, some protective friction, like CAPTCHAs, may indeed prove useful^{37–39}. Friction added to otherwise one-click sharing and liking will make the proliferation of both harmful and benign content more cumbersome and time-consuming. Friction is thought to prompt a more deliberate approach to sharing or liking content. Examples include exposing users to a contextual label³⁸, impeding the completion of an action with a prompt asking the user to reflect¹⁴, exacting micro-payments, or requiring users to spend mental resources through micro-exams such as quizzes and puzzles³⁵. Such friction strategies promise to deliver socio-political benefits by supporting cognitive autonomy while increasing the cognitive burden of sharing low-quality content³⁹.

Borrowing terminology from Tomalin³⁷, the type of friction of interest in this paper is *non-elective* for users—users cannot control exposure. This feature is also ruling *sludges* and *dark patterns*. Sludges generally refer to excessive friction, bad almost by definition and with a clear negative valence (e.g., bureaucratic form-filling). Dark patterns coerce, steer, or deceive people into making unintended and potentially harmful choices⁴⁰. To the contrary, the types of friction we consider in this paper aim to deliver social benefits. They are easily distinguishable from sludges and dark patterns as they are *overt*, neither *deceptive* nor *accidental*, but *intended*, *protective*, and *non-commercial*³⁷. Friction strategies sharing these characteristics may be *impeding* or *distracting*. For example, friction can impede an action by letting users complete it only after a micro-exam is passed. Certain nudges provide non-impeding friction; deliberation-promoting nudges, for example, are distracting but leave all options available to the user⁴⁰.

Recent research suggests that non-elective, overt, intended, protective friction is a promising tool to boost the accuracy and quality of information shared online. Adding as little friction as having users pause to think before sharing may prevent misinformation proliferation on social media: in a set of online experiments, participants who were asked to explain why a headline was true or false were less likely to share false information compared to control participants^{41,42}. In an effort to nudge users to consciously reflect on tweet content, Bhuiyan et al.⁴³ developed a browser extension that introduced a distracting emphasis on high-quality content and grayed out posts from low-quality sources. This raised the accuracy of tweet credibility assessments. Pennycook et al.^{14,15} see potential in reminding users of accuracy. They prompted experiment participants to rate the accuracy of a news headline before scrolling through an artificial social media news feed. Participants subsequently shared higher-quality content than a control group. The authors suggest to translate their findings into attention-based interventions subtly reminding users of accuracy to slow down the sharing of low-quality content online. In an experimental setting, Jahanbakhsh et al.⁴⁴ lend further support to the idea that interventions that nudge users to assess the accuracy of information as they share it yield overall lower sharing rates for false content over true content. Similarly, checking for accuracy assisted the fight against the illusory truth effect. This was demonstrated in a set of experiments that prompted participants to behave like fact checkers by asking them for initial truth ratings at first exposure⁴⁵.

We call this type of interventions *quality-recognition learning*, which may likewise be boosted by priming^{46,47}, testing, and retrieval^{48–50}. Learning may be further facilitated by memory boosters, such as repetitive interventions, as recently studied in the context of misinformation inoculation⁵¹. Learning through priming and nudging takes place without conscious

guidance. In contrast, testing and retrieval of previously absorbed knowledge take place consciously. Yet, both serve as learning events. Which mechanism takes precedence depends on the design of the friction prompt.

On Facebook, priming critical thinking made users less prone to trusting, liking, and sharing fake news about climate change⁴⁷. Reminding users of critical thinking was accomplished through considerations concerning news evaluation guidelines, using questions to help identify fake news as articulated by the Facebook Help Center (e.g., “Does the information in the post seem believable?”). Time pressure has further been shown to negatively influence the ability to distinguish true and false headlines⁵². Friction—as a means to reduce time pressure—may actively improve the discrimination of accurate from false information.

Most interventions by social media platforms to date do not impose restrictions on sharing⁵³. They tend to use redirection (suggesting content from authoritative sources such as the WHO during the COVID-19 pandemic) and content labeling (exposing users to additional context), thus preserving user choice and autonomy. More harsh but less common approaches include removing or down-ranking harmful content, (shadow) banning users who spread it, or decreasing its reach.

Various platforms have adopted friction interventions. Twitter implemented protective, non-elective friction that distracts or impedes. The platform introduced caps on automated tweeting³⁴ and a distracting label to pause users about to share state-affiliated media URLs⁵⁴. With limited success, they tested replacing retweets with “quote tweets,” requiring users to comment before they could share a post⁵⁵. Twitter also conducted a randomized controlled trial to curb offensive behavior, where users were asked to review replies in which harmful language was detected⁵⁶. Facebook has established policies to provide context labels from fact checkers. The platform reduces the distribution of, and engagement with, misinformation from repeat offenders by reducing the reach and visibility of their posts. While this intervention leads to a decrease in engagement with the offender in the short term, it can be compensated by an increase in the offender’s posts and followers. Furthermore, the limitation in reach can be reversed by deleting flagged posts⁵⁷. WhatsApp has taken first steps to counter the virality of misinformation by limiting the forwarding of messages to at most five contacts simultaneously⁵⁸. Instagram has introduced a distracting but non-impeding anti-bullying label that prompts users to pause by asking them “Are you sure you want to post this?”⁵⁹.

Unfortunately, reporting by social media platforms on the friction interventions deployed to date lacks transparency related to the testing and implementation process, making it difficult for researchers to study the different countermeasures. In addition, the effectiveness of content labeling has been challenged by research findings⁴⁵: while a common advice for dealing with fake news is to consider the source, people often struggle to remember sources⁶⁰. Tagging only some false news stories as “false” may boost the perceived accuracy of inaccurate but untagged stories due to an implied truth effect⁶¹.

A proposed field experiment

Let us discuss a concrete idea for a friction prompt and outline design avenues for a field experiment. Our main idea is to associate friction with learning through a prompt that quizzes users about community standards. This design aims to promote quality recognition before users may react to content. In an experimental environment, users would be faced with randomly assigned micro-exams (e.g., multiple choice) when they are about to share or like posts. The idea of impeding quizzes has most famously been established with CAPTCHA tests. Quizzes testing a reader’s understanding of an article have also been successfully used in the comment section of a public Norwegian broadcaster⁶². By involving questions about the platform’s governing community standards, the exams would help users further familiarize themselves with the rules put forth by the social platform in question⁵. Such a friction strategy falls into the *skill-adoption paradigm* of research on behavioral interventions, where participants learn the skills and strategies required to evaluate information quality⁶³. Lutzke et al.⁴⁷ showed that the combination of reminding users of critical thinking and questions

taken from community standards may help participants identify fake news. The display of community norms on Reddit increased compliance and prevented unruly and harassing conversations⁶⁴.

The proposed field experiment would measure whether engagement with low-quality posts drops when users are prompted with friction strategies incorporating quality-recognition learning based on community standards. Engagement with low-quality posts, measured by likes and shares, should be compared to engagement with high-quality content. A possible definition of low-quality posts could be based on violations of community standards. Such a field experiment may indeed contribute to research on testing effects^{48–50}, nudges, and priming^{14,47}. On the other hand, the experiment could elucidate the extent to which user beliefs, social identity, and social influence may interfere with abiding to learned norms.

In the proposed field experiment, participants will engage with posts as they would on their preferred platform. The friction intervention will take the form of a prompt during this process. Quizzes would randomly remind users of the platform's community standards (not only upon violating those, in contrast to Katsaros et al.⁵⁶). Quizzes may query the participants about, say, definitions, examples, and risks of violating misinformation or yet other community standards. The design is intended to promote fluency in the community standards rather than just awareness of particular sanctioning clauses. Since memory plays an important role in misinformation inoculations⁵¹, an interesting question to explore is whether the suggested random, repetitive reminders will serve as boosters to make the community standards more memorable.

Both intervention and control groups should be set up to test for causality⁶⁵, rather than exposing all study participants to the designed intervention as in Avram et al.²³. We also refer to Pennycook et al.⁶⁶ for a comprehensive guide on behavioral experiments related to misinformation and fake news.

Testing friction strategies in a real environment poses difficulties⁶⁷. Independent researchers do not have access to a platform such as X or Facebook to design and test real-time interventions in randomized controlled trials. One may alternatively carry out the experiment in a system designed to emulate a social media environment. Available options include Amazon Mechanical Turk^{41,68}, Volunteer Science⁶⁹, games such as Fakey⁷⁰, or open-source software such as the Mock Social Media Tool⁷¹. Each environment yields different levels of ecological validity, and experiments may be informed by empirical data about user activity and social network structure guiding online information sharing⁵⁸.

A key design decision is what to show participants in their feeds. The higher the desired ecological validity, the more sizeable the task of labeling posts to determine intervention effects. One extreme is to work solely with each user's own feed. While this yields the highest ecological validity, it comes with two downsides. First, one cannot be sure that users are exposed to misinformation at all, making the size of a sufficient data sample unknown. Second, all posts users have seen, or at least engaged with, must be labeled as misinformation or not. Labeling has either limited accuracy when derived from lists of low-credibility sources^{3,72}, or requires a lot of manual labor. Alternatively, one may work with synthetically curated feeds. A set of posts could be (partially) curated by researchers^{41,52,66,73–75}. These posts could be shown to all participants, thus ensuring that all users are shown the same misinformation posts while restricting the set of posts that must be labeled. Exposing participants to injected low-credibility content may raise valid ethical concerns, creating a potential need for mental health support during the study⁶⁶. The experimental design will have to weigh the advantages and limitations of both real and synthetic feeds.

As a policy, the friction strategy proposed here is non-elective, overt, protective, non-commercial, and impeding. We hypothesize that in the presence of learning, the needed impediment will be minimal. In contrast, the redirection and content labeling strategies that have dominated platform interventions are merely distracting. This allows harmful content to continue to circulate and places a greater burden on users to manage threats from such content themselves. Users who ignore labels or do not follow redirections remain vulnerable.

An important advantage of friction prompts is that they circumvent the task of identifying bad actors, which is resource-expensive and difficult^{2,3,67,76–78}. In contrast to other friction-based strategies^{43,56,57}, our approach does not rely on labeling of content as high- or low-quality by participants; one-click reactions may be interpreted as supporting the original post without invoking NLP techniques or human annotation to determine the tone of comments/replies. Friction prompts provide an unbiased, scalable strategy⁶³ that actively adds costs for inauthentic actors.

An agent-based model of friction in social media

We use an agent-based model of information sharing on social media to inform our experimental design. Our model is based on a version of *SimSoM*³⁵, amended by mechanisms for friction and learning. Further details about the model may be found in a technical report⁷⁹. In the model, posts are shared by agents, and appear on their news feeds. Posts represent content such as text, images, links, and hashtags. Each agent's news feed consists of posts shared by agents they follow. *SimSoM* assumes a directed follower network, similar to those on platforms like Bluesky, Mastodon, Threads, and Twitter/X.

At each time step, an agent is randomly selected to perform an action: post a new message with probability p (*post* scenario), or else re-share a post from their news feed (*share* scenario), as illustrated in Fig. 1. The new or re-shared post is added to the news feeds of the agent's followers. We use $p = 0.5$ based on empirical data for English-language tweets⁸⁰.

Each agent's feed contains the α most recent messages posted or re-shared by those they follow, i.e., their friends (Fig. 1). Although social media platforms use various algorithms to rank posts, reverse chronological order is a reasonable simplifying assumption because all platforms give high priority to recent posts. Other, more realistic ranking algorithms can be explored in future work³⁵. We use $\alpha = 15$ to represent the finite attention of the agents⁸¹, which gives rise to heavy-tailed distributions of post popularity and lifespan consistent with empirical data⁸².

Posts may vary in *quality* and in how *appealing* they are. Quality models an objective desirable property, such as accuracy or relevance of posts. Appeal models the quality of a post as *perceived* by agents. Both the quality q and the appeal a of a post are defined in the unit interval. We independently draw a post's quality and appeal from the normalized probability density functions $P(x) = \frac{(1-x)}{\int_0^1 (1-x)dx} = 2(1-x)$ with $x = \{q, a\}$.

These linear, independent distributions reflect the intuitions that high quality and high appeal do not necessarily coincide⁹ and that most content has low quality and low appeal.

Initially, an agent is able to discern the appeal of posts in their feed, but not their quality. Thus, we assume that the probability that an agent re-shares a post m is proportional to the post's appeal $a(m)$. While the appeal of a post does not change when it gets shared more often, a news feed may contain duplicates of the same message, e.g., if an agent follows two others sharing the same post. This further increases the chances that the message is re-shared, implicitly modeling an algorithmic bias that amplifies popular messages.

In our model, friction is restricted to the stage when agents re-share something posted by their friends (Fig. 1). In this *share* scenario, agents may face a friction prompt. This design follows the mechanism in the proposed experiment: friction triggers agents to pause, potentially impeding their sharing activity. The friction prompt forces agents to spend mental resources, for example taking a quiz. Agents may reconsider sharing the chosen post, or be prevented from sharing it if they fail to comply with the quiz. The probability that agents in the share scenario are exposed to friction and either reconsider or fail to comply is captured by the parameter f (see Fig. 1). The simulation records that an agent has been exposed to the friction prompt.

Agents may learn through exposure to the friction prompt, e.g., through an educational quiz or deliberation-triggering nudge reminding them to pay attention to quality. Learning in the model is kept at general level and allows to draw intuitions either from testing and retrieval or from

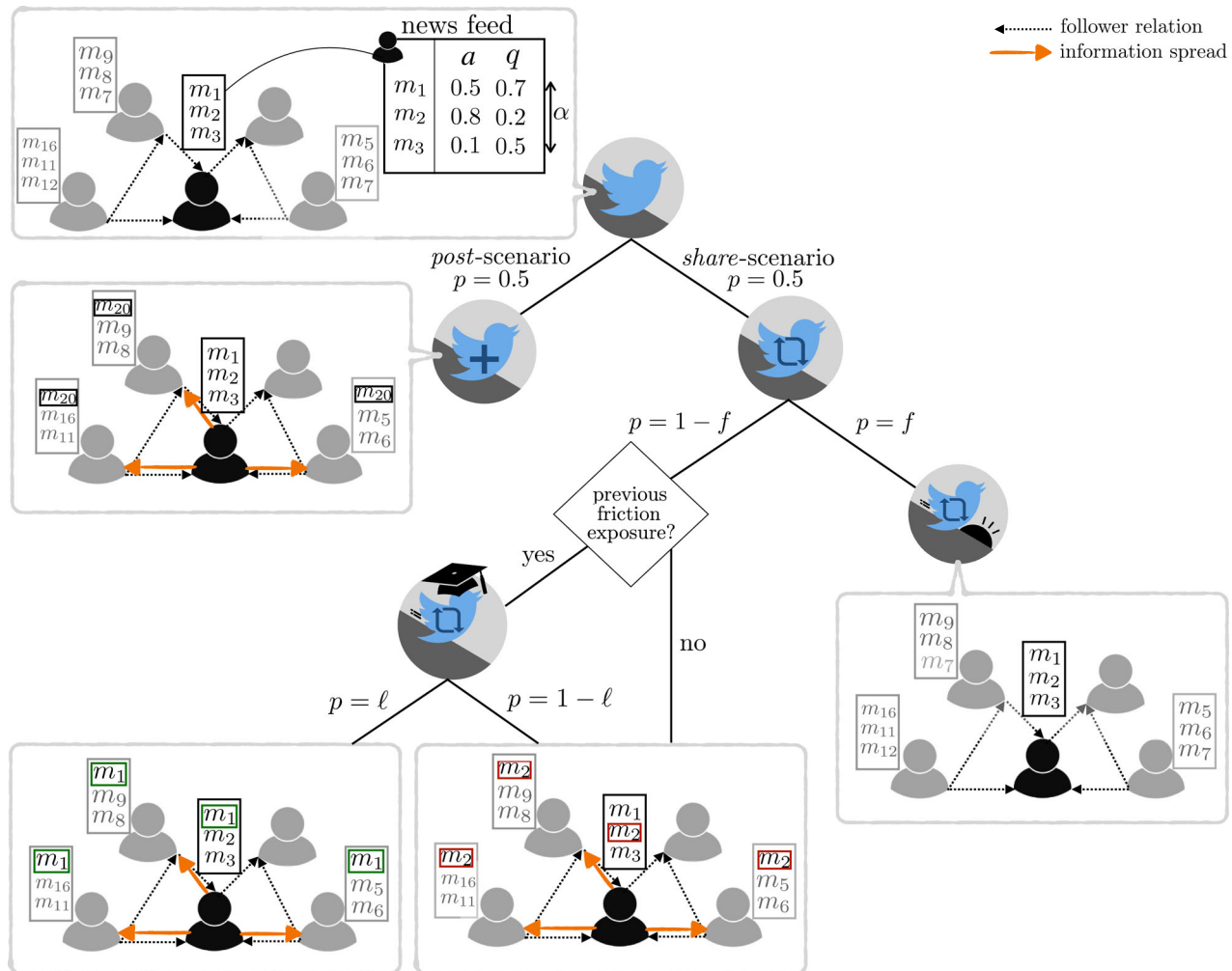


Fig. 1 | Information diffusion process. Each node has a news feed of size α , containing messages recently posted or re-shared by friends. The follower network is illustrated by dotted links pointing from an agent to their friends. Information spreads from agents to their followers, along the orange links (in the opposite direction of the follow links). The central black node represents an active agent posting a new message (here, m_{20}). The new message appears at the top of the followers' news feeds and the existing messages are shifted down. If α messages were

already present in a feed, the oldest one is discarded. If the agent wishes to re-share a message, with probability f friction prevents this. Else, with probability $1 - f$, the agent shares a post from their feed selected with probability proportional to the post's appeal (here, m_2)—unless the agent has been exposed to friction earlier and learned (with probability ℓ): in that case, the agent instead selects the post to share with probability proportional to the post's quality (here, m_1).

priming and nudging. The outcome—learning to recognize quality—does not change.

Agents who have previously been exposed to friction prompts may learn to discriminate between appeal and quality. However, agents are not always willing to learn or able to apply what they have learned; the parameter ℓ controls the probability with which agents learn to recognize quality after previous friction exposure. Formally, with probability ℓ , an agent previously exposed to a friction prompt selects a message m with probability proportional to its quality $q(m)$. With probability $1 - \ell$, the message to be shared is selected with probability proportional to its appeal $a(m)$, as agents may still make mistakes, forget, or be indifferent to quality.

To quantify the effect of friction and learning, our simulation tracks the changes in the overall quality of posts in the network—a desirable property of an online social network. The overall quality is calculated as the average quality $\hat{q}_T$ across all posts in the agent feeds, measured at the end of a simulation. Each simulation halts at convergence time T , when an exponential moving average of the overall quality stabilizes⁷⁹. We sweep values of the parameters f (induced friction) and ℓ (learning capability) in increments of 0.01 until 0.2 and increments of 0.1 thereafter. This yields 813 parameter combinations. For each of these combinations, we generate five follower networks with $N = 1000$ agents, mimicking realistic structural characteristics

like the presence of hubs and clustering. Finally we run 10 simulations for each network and average $\hat{q}_T$ across the $5 \times 10 = 50$ simulations.

Figure 2 shows that friction alone, in the absence of learning ($\ell = 0$), does not increase average quality. This is plausible, as friction is triggered independently of the quality or appeal of the post.

When friction-exposed agents learn, we observe a significant increase in average quality. The largest boost occurs when friction is small. In particular, when $f = 0.1$, the probability of an agent being prevented from re-sharing is $pf = 0.5 \cdot 0.1 = 0.05$. In this scenario, even a small measure of learning (e.g., $\ell \geq 0.1$) helps quality. However, higher learning probabilities boost quality further: for $\ell = 0.5$, $\hat{q}_T \approx 0.41$ —an increase of $\sim 17\%$ over the $\ell = 0$ baseline. And when learning occurs every time friction is applied ($\ell = 1$), we observe $\hat{q}_T \approx 0.47$ —an increase by more than one third.

Figure 2 also reveals that increasing the probability of friction further ($f > 0.1$) does not result in higher quality, no matter how much and how well agents learn. This finding suggests that while learning, only limited friction provides the best conditions for agents to apply learned behavior, even when accounting for forgetting, making mistakes, or intentional sharing of low-quality content ($\ell < 1$). The reason for quality to drop with high f is that agents cannot benefit from newly won awareness of post quality when they are prevented from sharing. There is a tradeoff between learning and friction

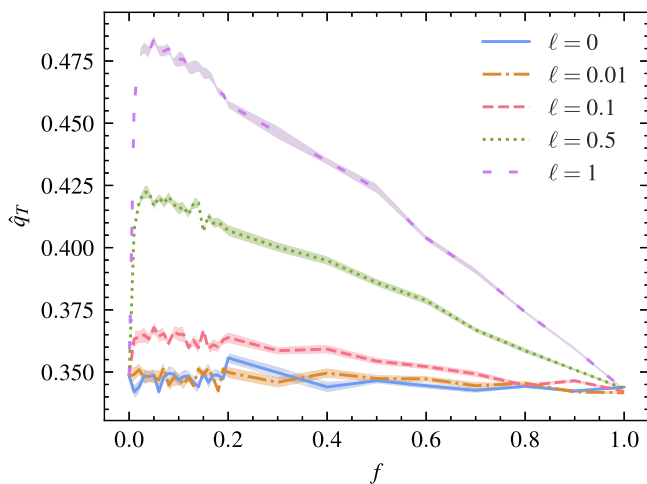


Fig. 2 | Average post quality $\hat{q}_T$ as a function of friction probability f , for different probabilities of learning ℓ . The subscript T indicates that average quality is measured at convergence. Shaded areas indicate standard errors.

in this model: when both ℓ and f are high, sharing would amplify high-quality content but it is too rare to counter the low-quality content that is still posted, leading to lower average quality. Therefore, our model suggests that combining rare friction prompts with quality-recognition learning is far more desirable than extensively restricting sharing.

Conclusion

This paper has proposed to experiment with friction and quality-recognition learning. The intervention may be easily deployed on social media platforms at scale as it does not require identifying low-quality content.

On the one hand, educating users on community standards will ease the burden of enforcing the standards—especially now when many social platforms are dialing back content moderation⁸³. On the other hand, users who are well-versed in community standards may encourage platforms to enforce their content moderation rules transparently and consistently. Hopefully, learning about and enforcing community standards will stimulate public debate on democracy, freedom of expression, privacy, accountability, and user rights on online public spaces—principles often affirmed by tech giants themselves.

The preliminary results from our agent-based model suggest that a small amount of friction combined with quality-recognition learning can significantly increase the quality of content on a social media platform. The average quality increases with just a bit of friction, even with far-from-perfect learning. On the other hand, excessive friction, even with perfect learning, has a negative effect on quality as agents are prevented from applying their newly acquired knowledge. A very low friction probability would also have the desirable effect of making the proposed intervention minimally invasive in an actual social media environment.

These predictions are based on assumptions implicit in our model. For example, the model focuses on one-click sharing, thus neglecting that learning could positively affect the quality of original posts. Despite such limitations, the model's predictions provide us with specific experimental design choices to be tested in the field.

All large social media platforms should have a keen interest in interventions improving information quality—especially when the quality increase is based on user familiarity with the community standards of the platform in question. However, certain platforms, such as X and Facebook, have recently moved away from moderating misinformation. Open platforms, such as Bluesky and Mastodon, may be best positioned to roll out experiments with scalable friction strategies. We encourage any platform undertaking such interventions to share data and tools with researchers, as required by the Digital Services Act of the European Union⁸⁴, so they may aid in assessing intervention efficacy.

Data availability

The code for the agent-based model and to reproduce the analysis is accessible on GitHub (<https://github.com/LJ-9/Friction-Social-Media-Model>)⁸⁵.

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Author contributions

All authors conceived and designed the model and simulations. L.J. performed the simulations. All authors analyzed the data, contributed materials and analysis tools, and wrote the paper.

Competing interests

The authors declare no competing interests.

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